

# Maximising Edge Density in Graphs Subject to Connectivity Constraints

**Erdős Problem 915, from Proof to Search**

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Maximising edge density in graphs  
under connectivity constraints

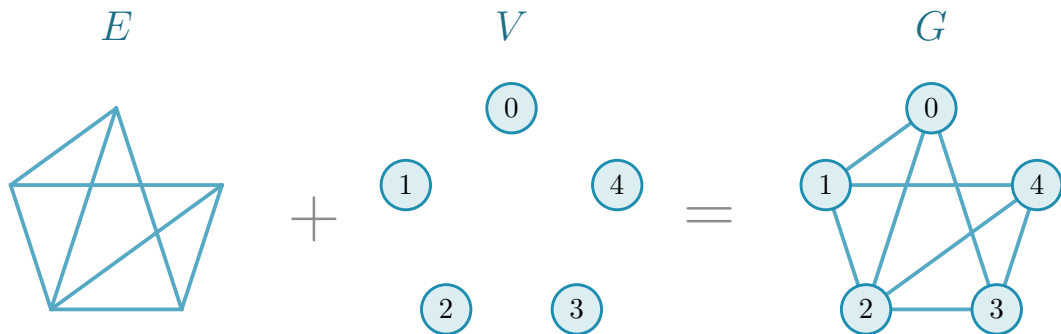


*translation*

networks with **many links** and **low connectivity**

# Graph

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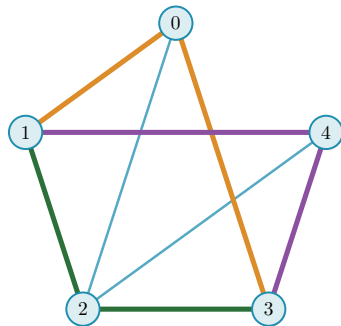


# Erdős Problem 915

$$\ell_m(n) = \max_{|V(G)|=n} \left\{ |E(G)| : \lambda_G^{\max} \leq m-1 \right\}$$

$$\lambda_G^{\max} = \max_{u \neq v} \lambda_G(u, v)$$

$\lambda_G(u, v)$  = the largest number of  $u-v$  routes that share **no edge**: one colour each

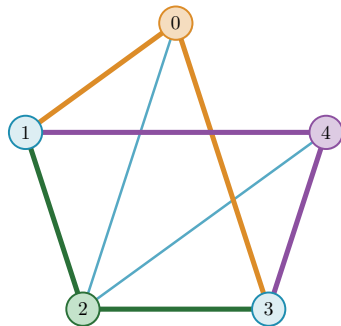


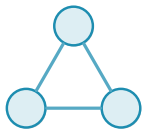
# The same question, vertex by vertex

$$k_m(n) = \max_{|V(G)|=n} \left\{ |E(G)| : \kappa_G^{\max} \leq m-1 \right\}$$

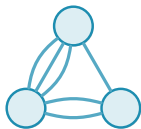
$$\kappa_G^{\max} = \max_{u \neq v} \kappa_G(u, v)$$

$\kappa_G(u, v)$  = the largest number of  $u$ - $v$  routes that share **no interior vertex**

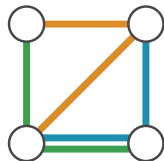




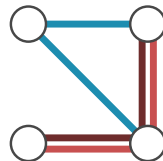
*simple graph*



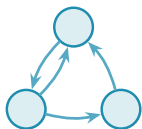
*multigraph*



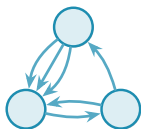
*hypergraph*



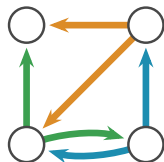
*multihypergraph*



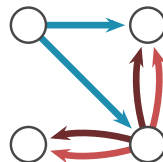
*simple digraph*



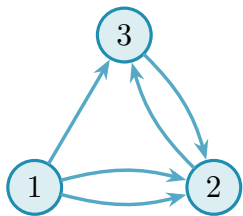
*directed multigraph*



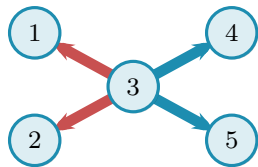
*directed hypergraph*



*directed multihypergraph*



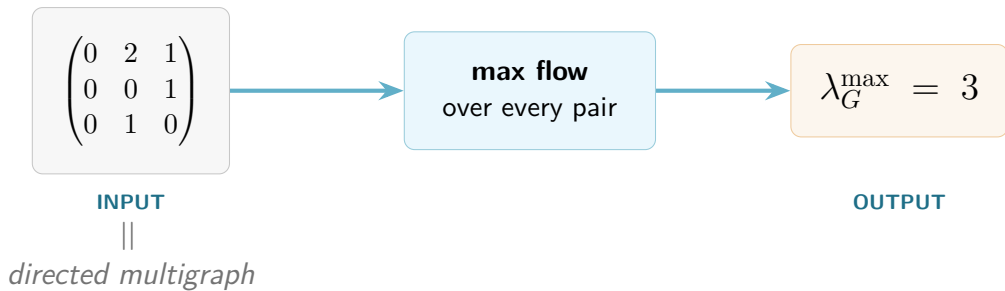
$$\begin{pmatrix} \mathbf{0} & 2 & 1 \\ 0 & \mathbf{0} & 1 \\ 0 & 1 & \mathbf{0} \end{pmatrix}$$



list of hyperedges:

$$e_1 : \{3 \mid 1, 2\}$$

$$e_2 : \{3 \mid 4, 5\}$$

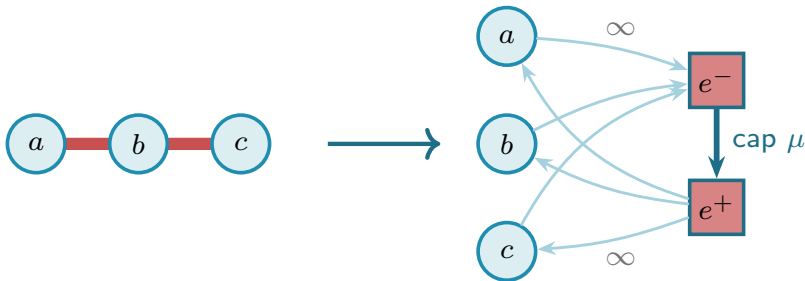




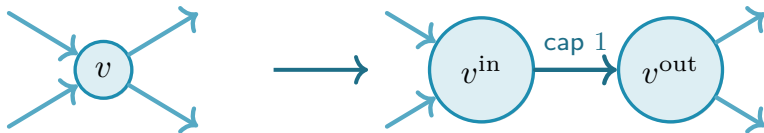
undirected  $\rightarrow$  directed



# hypergraph $\rightarrow$ non-hypergraph



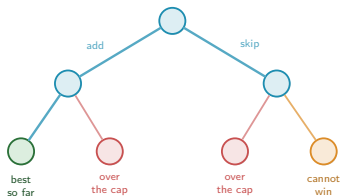
$$\kappa \rightarrow \lambda$$



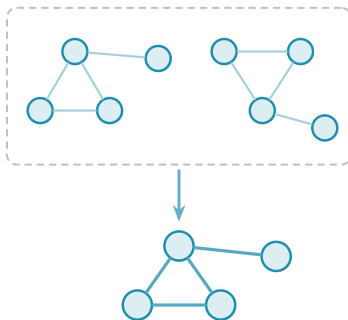
every intermediate vertex is split this way  
the flow itself runs from  $s^{\text{out}}$  to  $t^{\text{in}}$

# Efficiency: three ways to make the sweep finish

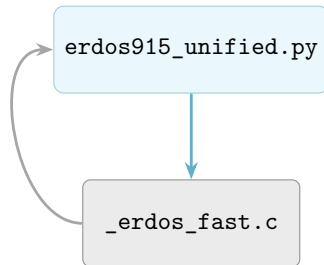
## pruning



## generating search

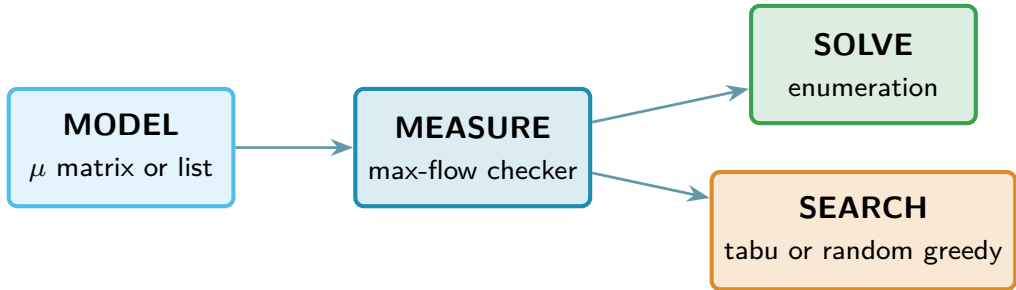


## C: faster max flow



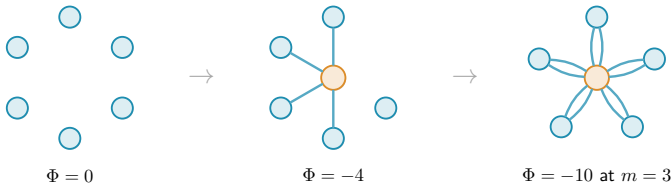
# Solving versus searching

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# The search: one number per graph

$$\Phi(G) = \underbrace{-|E(G)|}_{\substack{\text{reward} \\ \text{more edges, lower energy}}} + \underbrace{\Lambda \cdot \max(0, c^{\max}(G) - (m - 1))}_{\substack{\text{penalty} \\ \text{discourages excess connectivity}}}$$

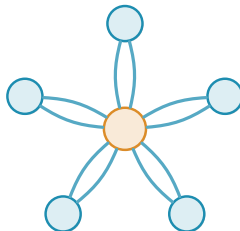


# Extremal families the search rediscovers



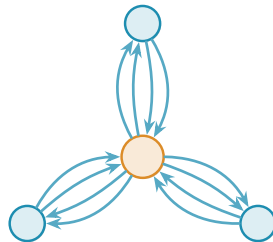
*tree*

$$n = 6, \ell_2(n) = 5$$



*multigraph star*

$$n = 6, m = 3, L_3(n) = 10$$

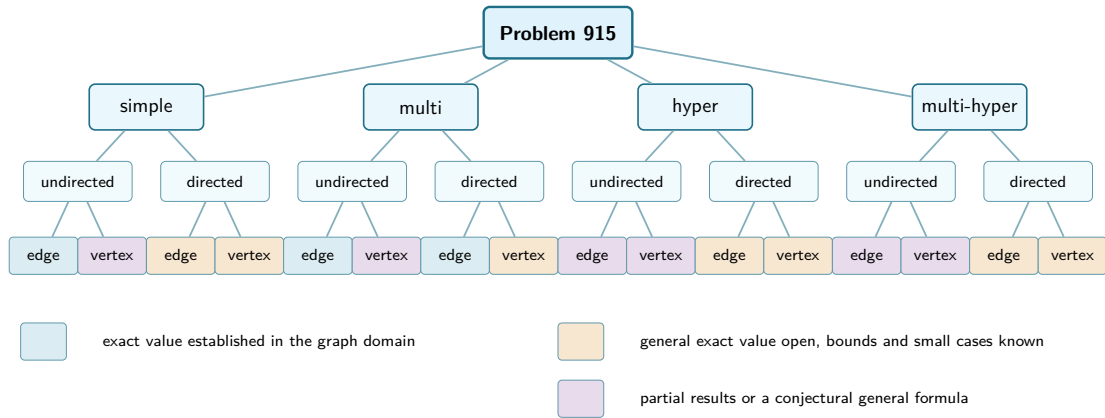


*double star*

$$n = 4, m = 3, L_3^{\text{dir}}(n) = 12$$

examples of the families, not the saved search witnesses

# Status of the sixteen variants





# Thank you

`github.com/louisvandenbruwaene/thesis`

# Backup: the sixteen variants

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## WHAT IS KNOWN ABOUT A BOUND

PROVED

established, with a proof the author has checked

PROPOSED

conjecture with a proposed proof the author has not checked line by line

OPEN

no general value established

## WHAT A CONSTRUCTION IS WORTH

EXTREMAL

attains the maximum in the stated range

NEAR-EXTREMAL

the densest family reached here, with no matching upper bound

ASYMPTOTICALLY EXTREMAL

matches the leading term of the upper bound

# Simple graph · edge

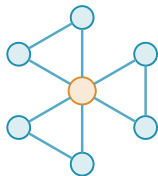
$$\ell_m(n)$$

**UPPER**  $\left\lfloor \frac{m(n-1)}{2} \right\rfloor$  **PROVED** Mader, via Gomory–Hu

**LOWER**  $\left\lfloor \frac{m(n-1)}{2} \right\rfloor$  **PROVED** for every  $n \geq m$

- the two meet for  $n \geq m$ , so the value is exact there
- for  $n < m$  it is  $\binom{n}{2}$ : not even  $K_n$  carries  $m$  routes

**COMPUTED** Tabu search returns the tree at  $n = 6, m = 2$ :  $5 = \ell_2(6)$ .



hub-and-spokes **EXTREMAL**

# Simple graph · vertex

$$k_m(n)$$

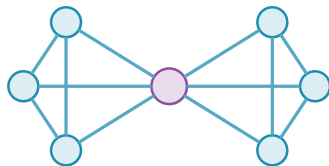
**UPPER**  $k_m(n) = \ell_m(n)$  for  $m \leq 4$ ,  $k_5(n) = \lfloor \frac{8n}{3} \rfloor - 4$  for  $n \geq 14$  **PROVED**

$m \geq 6$ : **OPEN**

**LOWER**  $c_m > m/2$  for  $m \geq 5$  **PROPOSED** under the block formulation

- at  $m = 5$  the two agree to  $n = 13$ , first gap 33 against 32 at  $n = 14$
- $c_m = \lim_n k_m(n)/(n - 1)$ , and  $m/2 < c_m \leq 2(m - 1)$  for  $m \geq 5$

**COMPUTED** Block sweeps reach small blocks only, and the recursive constructions in the literature show why they cannot fix  $c_m$ .



2-connected blocks at a cut vertex **NEAR-EXTREMAL**

# Simple digraph · arc

$$\ell_m^{\text{dir}}(n)$$

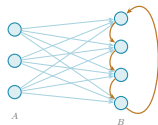
**UPPER**  $M(n) = \max\left(2(n-1), \left\lfloor \frac{n^2}{4} \right\rfloor\right)$  at  $m = 2$  **PROVED**

$\left\lfloor \frac{n^2}{4} \right\rfloor + 4(m-1)(n-1)$  at every  $n$  **PROVED**

**LOWER**  $\frac{n^2}{4} + \frac{(m-2)n}{2} + O_m(1)$  **PROVED**

- for fixed  $m \geq 3$ ,  $\ell_m^{\text{dir}}(n) = n^2/4 + \Theta_m(n)$ ; the linear coefficient is open
- this family refutes the first guess  $\max(m(n-1), \left\lfloor \frac{n^2}{4} \right\rfloor)$  at  $m = 3$ ,  $n = 10$

**COMPUTED** Enumeration confirms  $m = 2$  through  $n = 7$ . Tabu returns 6, 10 and 12 at  $n = 4$ , 6 and 7.



augmented bipartite digraph

**ASYMPTOTICALLY EXTREMAL**

# Simple digraph · vertex

$$k_m^{\text{dir}}(n)$$

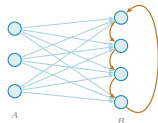
**UPPER**  $M(n)$  at  $m = 2$  **PROVED**

$$\frac{n^2}{4} + \Theta_m(n) \text{ for fixed } m \geq 3 \quad \textbf{PROVED}$$

**LOWER** the same bipartite family **PROVED**

- equality with the arc value at  $m \geq 3$  is open: an arc upper bound does not constrain the vertex-feasible family, which contains it

**COMPUTED** Enumeration confirms  $m = 2$  through  $n = 7$ , and the two separations agree in the completed  $m = 3$  runs through  $n = 6$ .



augmented bipartite digraph

**ASYMPTOTICALLY EXTREMAL**

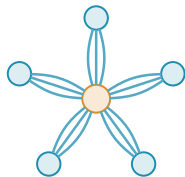
# Multigraph · edge

$$L_m(n)$$

**UPPER**  $(m-1)(n-1)$  **PROVED** a cut count, any shape

**LOWER**  $(m-1)(n-1)$  **PROVED**

- exact for every  $n$  and  $m$
- multiplicity removes the simple graph's distance-one restriction



tree, every edge repeated  
 $m-1$  times **EXTREMAL**

**COMPUTED** Tabu search returns the multiplicity-2 star at  $n=6$ ,  $m=3$ :  $10 = L_3(6)$ .

$$K_m(n)$$

**UPPER**  $K_2(n) = n - 1$ ,  $K_3(n) = 2(n - 1)$  **PROVED**

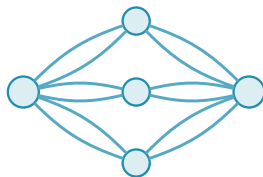
$$K_m(n) \leq (2 + o(1)) m^2 n$$
 **PROPOSED**

**LOWER**  $K_3(n) \geq 2(n - 1)$  **PROVED**

$$K_m(n) \geq (3 - 2\sqrt{2} + o(1)) m^2 n$$
 **PROPOSED**

- the two coefficients bracket the growth as  $m \rightarrow \infty$  with  $n/m \rightarrow \infty$ , and no common limit is established
- parallel copies are separated routes, so this is not the simple problem:  $K_5(4) = 14$  against  $L_5(4) = 12$

**COMPUTED**  $K_5(4) = 14$  and  $K_5(5) = 19$  by exhaustion. A nine-vertex sweep over all 191826 two-connected graphs with at least fourteen edges gives  $g_6(9) = 54$ .



thickened complete bipartite  
block **NEAR-EXTREMAL**



# Directed multigraph · arc

$$L_m^{\text{dir}}(n)$$

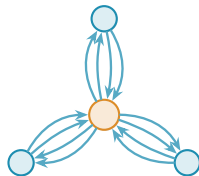
**UPPER**  $(m-1)M(n)$  **PROVED** a sparse skeleton per unit of  $m-1$

**LOWER**  $(m-1)M(n)$  **PROVED** two branches, both attained

- exact for every  $n$  and  $m$ , and the fractional optimum is  $M(n)$  too, so one  $m$ -free optimisation bounds every threshold
- the classification of the extremisers for  $n \geq \max(8, 2m)$  stays

**PROPOSED**

**COMPUTED** Tabu returns the double star at  $m=3$ : 12 at  $n=4$  and 16 at  $n=5$ .



thickened bidirected tree

**EXTREMAL**

linear branch,  $n \leq 7$

# Directed multigraph · vertex

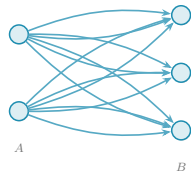
$$K_m^{\text{dir}}(n)$$

**UPPER**  $(m-1)\frac{n^2}{4} + O_m(n)$  **PROVED**

**LOWER**  $(m-1)\lfloor n^2/4 \rfloor$ , thickened bipartite **PROVED**

- the arc extremiser is not even a finite optimum here:  
 $K_6^{\text{dir}}(3) = 24$  against an established arc maximum of 20
- the exact correction and the best block shapes are open

**COMPUTED** Direct enumeration gives  $K_6^{\text{dir}}(3) = 24$ .



thickened one-way bipartite  
digraph

**ASYMPTOTICALLY EXTREMAL**

# Simple hypergraph · edge

$$\ell_m^{(r)}(n)$$

**UPPER**  $\left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor$  **PROVED** induction on a hyperedge cut

**LOWER** the same value **PROVED** when  $m-1 \leq \binom{n-2}{r-2}$

- one hyperedge spans  $r$  vertices but serves only one route
- outside that range attainment is not guaranteed: at  $n = r = m = 3$  the bound reads 2 and only one hyperedge exists

**COMPUTED** Exhaustive enumeration gives 11 at  $(n, m, r) = (6, 6, 3)$ , below the bound 12.



star hypertree ( $m = 2$ )

**EXTREMAL**

# Simple hypergraph · vertex

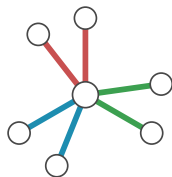
$$k_m^{(r)}(n)$$

**UPPER**  $\left\lfloor \frac{n-1}{r-1} \right\rfloor$  at  $m = 2$ ,  $\left\lfloor \frac{2(n-1)}{r-1} \right\rfloor$  at  $m = 3$  **PROVED**  
 $m \geq 4$ : **OPEN**

**LOWER** star at  $m = 2$ , degree-capped hub at  $m = 3$  **PROVED**  
 $m = 3$  for  $2 < r < n$

- $m = 2$  is feasible exactly when the incidence graph is a forest
- $m = 3$  comes from the incidence-rank inequality
- an edge bound does not transfer:  $\kappa \leq \lambda$  points the wrong way, vertex feasibility being the weaker restriction

**COMPUTED** Enumeration at  $m = 3$ ,  $r = 3$  gives 1, 3, 4 for  $n = 3, 4, 5$ : the bound  $\lfloor 2(n-1)/(r-1) \rfloor$  from  $n = 4$  on, one short at  $n = r = 3$ .



star hypertree ( $m = 2$ )

**EXTREMAL**

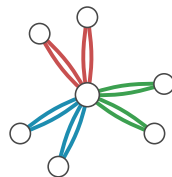
# Multihypergraph · edge

$$L_m^{(r)}(n)$$

**UPPER**  $\left\lfloor \frac{(m-1)(n-1)}{r-1} \right\rfloor$  **PROVED** unchanged by repetition

**LOWER**  $\frac{(m-1)(n-1)}{r-1}$  **PROVED** when  $(r-1) \mid (n-1)$

- the constructions change even though the bound does not
- at  $n = r = m = 3$ , two copies of the only triple reach a value the simple model cannot



star hypertree, edges repeated  
 $m-1$  times **EXTREMAL**

**COMPUTED** A twelve-copy construction at  $(n, m, r) = (6, 6, 3)$  exceeds the simple edge optimum of 11.

# Multihypergraph · vertex

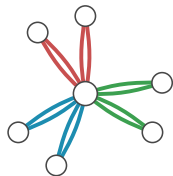
$$K_m^{(r)}(n)$$

**UPPER**  $\left\lfloor \frac{n-1}{r-1} \right\rfloor$  at  $m = 2$ ,  $\left\lfloor \frac{2(n-1)}{r-1} \right\rfloor$  at  $m = 3$  **PROVED**  
larger  $m$ : **OPEN**

**LOWER** the repeated star adds attainment cases at  $m = 3$   
**PROVED**

- repetition does not help at  $m = 2$ : two copies of one hyperedge are already two separated routes

**COMPUTED** The twelve-copy edge witness at  $(6, 6, 3)$  is a vertex lower bound, and does not prove the vertex maximum is twelve. A sweep at  $m = 4$  attains  $\lfloor 3(n-1)/(r-1) \rfloor$  and excludes one more in eleven multihypergraph cases, all  $r \geq 3$ .



repeated star hypertree

**EXTREMAL**  $m \leq 3$ , and  
 $(r-1) \mid (n-1)$  at  $m = 3$

# Directed simple hypergraph · arc

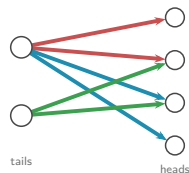
$$\ell_m^{\text{dir}(r)}(n)$$

**UPPER**  $(1 + o(1)) \frac{(m-1)n^2}{4(r-1)}$  **PROVED** shadow digraph

**LOWER** the same constant **PROVED** two-layer,  $r \geq 3$

- the two meet, so the leading term is settled and the finite exact values are **OPEN**
- the matching argument covers distinct two-step routes sharing one hyperedge

**COMPUTED** At  $n = m = 6$ ,  $r = 3$ , five distinct outgoing hyperedges per tail give 30, with a direct degree-cut feasibility check.



two-layer construction

**ASYMPTOTICALLY EXTREMAL**

# Directed simple hypergraph · vertex

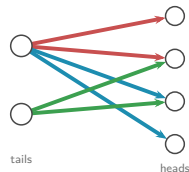
$$k_m^{\text{dir}(r)}(n)$$

**UPPER**  $(1 + o(1)) \frac{(m-1)n^2}{4(r-1)}$  **PROVED**

**LOWER** the same constant **PROVED** two-layer,  $r \geq 3$ , fixed  $m$  and  $r$   
as  $n \rightarrow \infty$

- the shadow argument re-runs under the stricter separation, so the leading term is the same
- whether the finite edge and vertex maxima agree is **OPEN**

**COMPUTED** Random greedy growth with shuffled restarts measures what the implemented search finds, and establishes no optimum.



two-layer construction

**ASYMPTOTICALLY EXTREMAL**



# Directed multihypergraph · arc

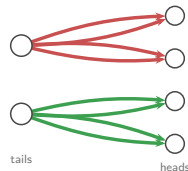
$$L_m^{\text{dir}(r)}(n)$$

**UPPER**  $(1 + o(1)) \frac{(m-1)n^2}{4(r-1)}$  **PROVED** forward, backward and general alike

**LOWER** the same constant **PROVED** repeats lift the head-set limit

- the finite values can differ from the simple model, and are **OPEN**
- at  $r = 2$  these are directed multigraphs, where the simple coefficient is  $1/4$  instead, which is why  $r \geq 3$  is kept

**COMPUTED** The plotted experiments use rank three and the forward convention. General orientations are checked separately in the appendix.



two-layer, hyperedges repeated

**ASYMPTOTICALLY EXTREMAL**

# Directed multihypergraph · vertex

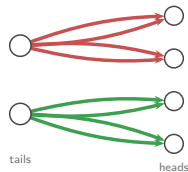
$$K_m^{\text{dir}(r)}(n)$$

**UPPER**  $(1 + o(1)) \frac{(m-1)n^2}{4(r-1)}$  **PROVED**

**LOWER** the same constant **PROVED**

- the same leading term under both separations
- finite exact values **OPEN**, and whether forward and general agree once  $n > r$  is open too

**COMPUTED** The rank-three forward-convention runs are not an exhaustive comparison of the three orientation models.



two-layer, hyperedges repeated

**ASYMPTOTICALLY EXTREMAL**